1. Integration

1.1. Introduction

- The integral can be thought of as the anti-derivative, or the reverse of differentiation.

In mathematical terms

$$\text{If} \quad \frac{d}{dx}F(x) = f(x), \quad \text{then} \quad \int f(x), dx = F(x) + C$$

Where:

- c is an **arbitrary constant** with no specific value.
  - This can be thought of as the value lost due to the differentiation of a constant number.
- dx represents that the function is being integrated with respect to x.

Basics of Indefinite Integration

Reverse power rule

$$\int x^n, dx = \frac{x^{n+1}}{n+1} + c \quad \text{for any real number } n \neq -1$$

Sum and Difference rule

$$\int f(x) \pm g(x), dx = \int f(x), dx \pm \int g(x), dx$$

Scalar multiple rule

$$\int a \cdot f(x), dx = a \int f(x), dx \quad \text{where } a \text{ is a constant number}$$

Constant rule

$$\int a, dx = ax + c \quad \text{where } a \text{ is a constant number}$$

1.2. Reverse Chain Rule

- The reverse chain rule helps us integrate **linear functions** that are raised to an exponent.

Consider $g(f(x))$ such that

$$\frac{d}{dx}g(\ f(x)\ ) = g\prime(\ f(x)\ ) \times f\prime(x)$$

We can integrate the result as such

$$\int g\prime(\ f(x)\ ) \times f\prime(x)\tilde{d}x = g(\ f(x)\ ) + c$$

An important application of the reverse chain rule is shown below.
[a and b are constant real numbers]

$$\int (ax+b)^n, dx = \frac{(ax+b)^{n+1}}{\left[\frac{d}{dx}(ax+b)\right] \times (n+1)} + C$$

This simplifies to

$$\int (ax+b)^n, dx = \frac{(ax+b)^{n+1}}{a \times (n+1)} + C$$

- This is valid for all real numbers $n \neq -1, 0$

Example

Integrate $\frac{1}{(2x+3)^2}$.

Answer

Using the reverse chain rule

$$\int (2x+3)^{-2} = \frac{(2x+3)^{-2+1}}{\left[\frac{d}{dx}(2x+3)\right] \times (-2+1)} + c$$

This simplifies to

$$\frac{1}{4x+6} + c$$

1.3. Finding the arbitrary constant

- To be able to find the value of the constant we must be given a point on the curve.

Consider a point (e, f) on the curve $G(x)$, where $G(x) = F(x) + c$.
We are given a derivative $f(x)$. To find the original equation:

$$\int f(x)\tilde{d}x = F(x) + c$$

Using the points given

$$f = F(e) + c \implies c = f - F(e)$$

Example

A curve is such that $\frac{dy}{dx} = 2x^2 - 5$. Given that the point $(3, 8)$ lies on the curve, find the equation of the curve.

Answer

We begin by integrating the derivative given.

$$\int 2x^2 - 5\tilde{d}x = \frac{2x^{2+1}}{2+1} - 5x + c$$

Giving us

$$y = \frac{2x^3}{3} - 5x + c$$

We can now substitute in the points given in the question

$$8 = \frac{2(3)^3}{3} - 5(3) + c \implies c = +5$$

So the equation of the curve is

$$y = \frac{2x^3}{3} - 5x + 5$$

1.4. Definite Integration

- A definite integral is an integral that is described by an upper and lower limit.

Fundamental theorem of Calculus

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$$\int_{x=b}^{x=a} f(x)\,dx = [F(x)]_b^a = F(a) - F(b) \quad [\text{Where } b \leq x \leq a]$$

Where:

- a is the upper limit.
- b is the lower limit.
- Note that adding the arbitrary constant is not necessary in definite integration as it will get cancelled out.

Basics of Definite Integration

- The upper and lower limit **must be the same** for this rule to apply

Sum and Difference rule

$$\int_b^a f(x) \pm g(x)\,dx = \int_b^a f(x) \pm \int_b^a g(x)$$

Scalar multiple rule

$$\int_b^a zf(x)\,dx = z \times \int_b^a f(x)\,dx \text{ [Where } z \text{ is a constant number]}$$

Change of limits rule

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$$

We also come across some useful results

$$\int_a^a f(x)\,dx = 0$$

$$\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx \quad [\text{Where } a < b < c \text{ and } f(x) > 0]$$

Example

Compute the value of $\int_2^4 3x^3 - 4x^2 + 2x + 5\,dx$.

Answer

We first start by integrating.

$$\int_2^4 3x^3 - 4x^2 + 2x + 5\,dx = \left(\frac{3x^{3+1}}{3+1} - \frac{4x^{2+1}}{2+1} + \frac{2x^{1+1}}{1+1} + 5x\right)_2^4$$

This simplifies to

$$\left(\frac{3x^4}{4} - \frac{4x^4}{4} + x^2 + 5x\right)_2^4$$

Now we can substitute in the upper and lower limit as such

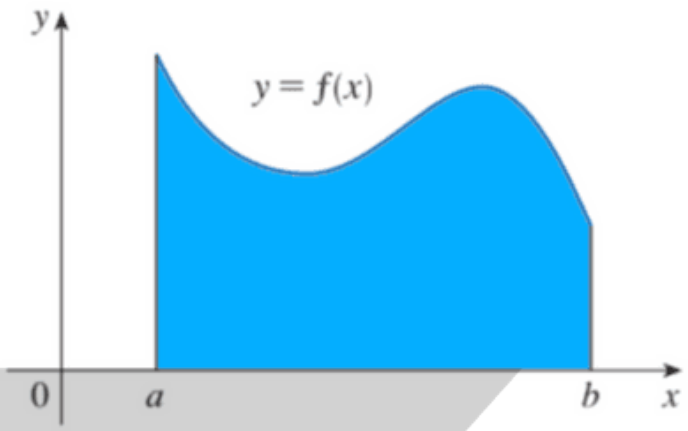
$$\left(\frac{3(4)^4}{4} - \frac{4(4)^4}{4} + (4)^2 + (4)x\right) - \left(\frac{3(2)^4}{4} - \frac{4(2)^4}{4} + (2)^2 + 5(2)\right)$$

$$\left(\frac{428}{3}\right) - \left(\frac{46}{3}\right) = \frac{382}{3}$$

$$\int_2^4 3x^3 - 4x^2 + 2x + 5\,dx = \frac{382}{3}$$

1.5. Area bounded by the x-axis

Finding the area under a curve is an application of definite integration.



- The definite integral below gives us the area under the graph bounded by the x-axis.

$$\int_{x=a}^{x=b} f(x)\,dx = F(b) - F(a) \quad [\text{Where } a \leq x \leq b]$$

- Here $f(x)$ must be in terms of x.
- Integrating a constant, such as $y = c$, with limits a and b will give you the area of a rectangle, bounded by the x-axis, with length c and breadth $b - a$.

The area bounded below the x-axis

- The area bounded below the x-axis is negative, but the **positive value** must be taken as area can never have a negative value.

So the area under the x-axis can be written as

$$-\int_a^b f(x)\,dx$$

To find the area under a graph as such:

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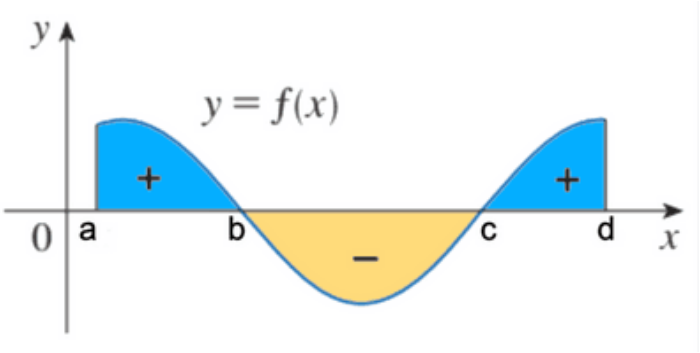


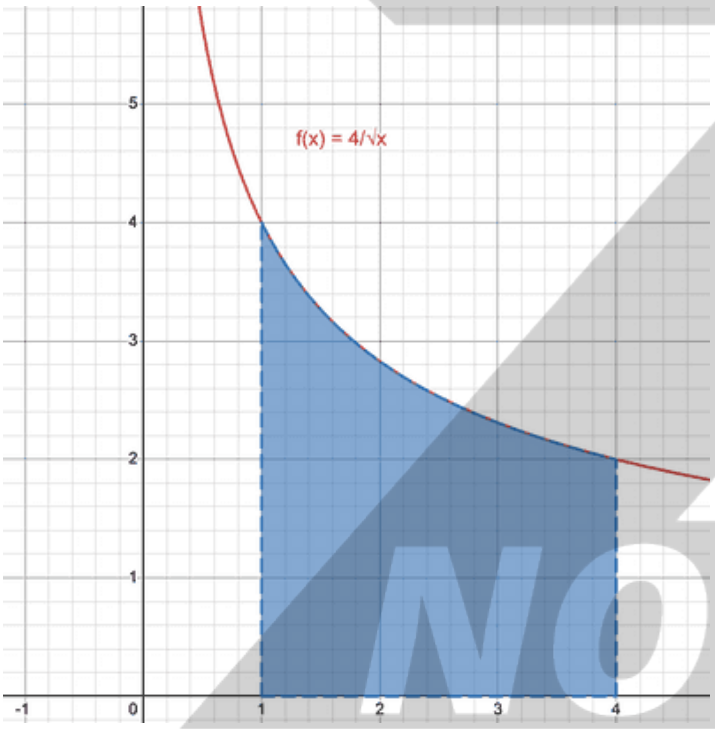
We can use 3 different definite integrals

$$\int_a^b f(x)\,dx - \int_b^c f(x)\,dx + \int_c^d f(x)\,dx$$

Example

A curve has the equation

$$f(x) = \frac{4}{\sqrt{x}}$$



Find the area of the shaded region enclosed by the curve, the x -axis and the lines $x = 1$ and $x = 4$.

Answer

From the question

- $x = 4$ is the upper limit as it is the greater number.
- $x = 1$ is the lower limit as it is the smaller number.

Now we can integrate the function $f(x)$

$$\int_1^4 4x^{-\frac{1}{2}}\,dx$$

Upon integrating, we get

$$\frac{4x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} = 8x^{\frac{1}{2}}$$

Now we can substitute in our limits

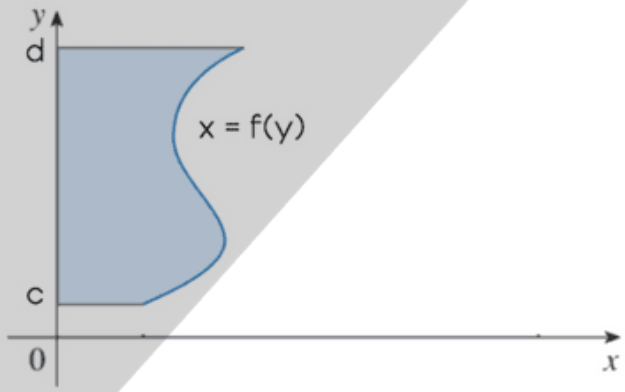
$$[8x^{\frac{1}{2}}]_1^4$$

$$\left(8(4)^{\frac{1}{2}}\right) - \left(8(1)^{\frac{1}{2}}\right) = 8$$

So the area of the shaded region is 8.

1.6. Area bounded by the y-axis

- To find the area bounded by the y-axis, we need to get an equation in terms of y, such as $x = F(y)$.
  - The limits must converted to y-values, if required.

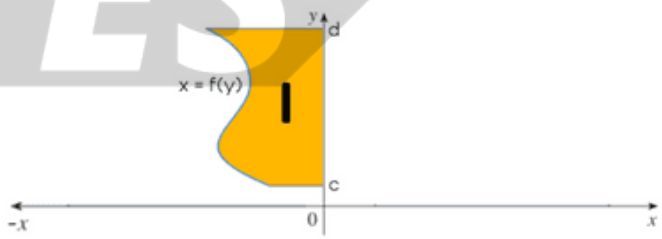


- The definite integral below gives us the area bounded by the y-axis.

$$\int_{y=c}^{y=d} f(y), dy = F(d) - F(c) \quad [\text{where} \quad c \leq y \leq d]$$

Area bounded left of the y-axis

- The area bounded below the y-axis is negative, but the **positive value ** must be taken as area can never have a negative value



So the area left of the y-axis can be written as

$$-\int_c^d f(y)\tilde{d}y$$

Example

A curve has the equation

$$y = 2 + \sqrt{x+3}$$

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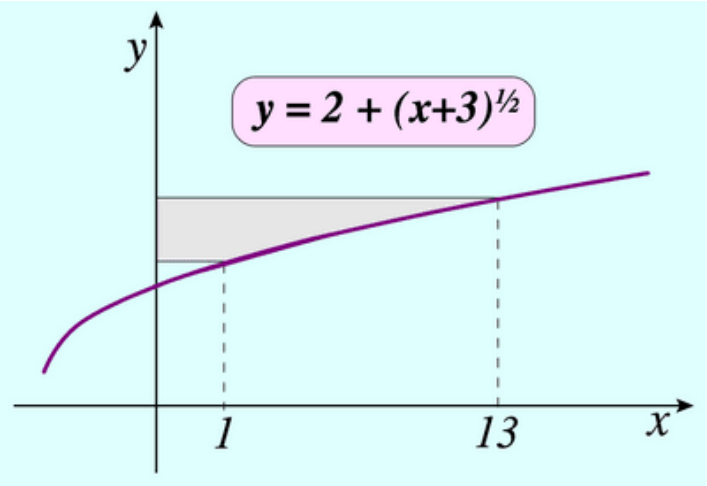


Find the area of the shaded region, bounded by the y-axis and the curve.

Answer

We start by making x the subject of the formula

$$x = (y-2)^2 - 3 = y^2 - 4y + 1$$

Next, find the corresponding y-values for the x limits.

$$y = 2 + \sqrt{1+3} = 4$$

$$y = 2 + \sqrt{13+3} = 6$$

We can now begin by integrating, along with the upper and lower limit

$$\int_4^6 y^2 - 4y + 1 \tilde{d}y = \left(\frac{y^3}{3} - \frac{4y^2}{2} + y\right)_4^6$$

Substituting in the limits gives us

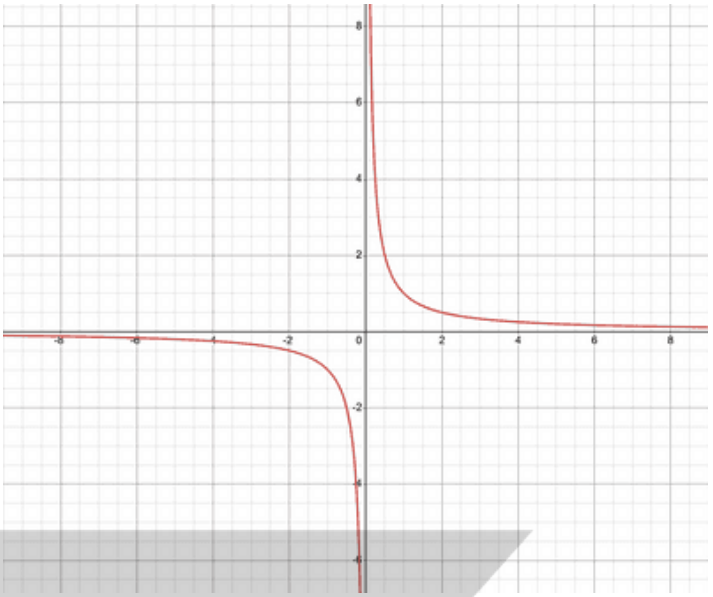
$$\left(\frac{(4)^3}{3} - 2(4)^2 + 4\right) - \left(\frac{(6)^3}{3} - 2(6)^2 + 6\right) = \frac{38}{3}$$

So the area of the shaded region is $\frac{38}{3}$.

1.7. Improper Integration

- Improper integration is a type of definite integration where the function being integrated is undefined at the limits given.

Consider the graph of the function $f(x) = \frac{1}{x}$



Here:

- As x approaches 0, $f(x)$ approaches ∞
  - This is written as $x \to 0$, $f(x) \to \infty$
- As x approaches ∞ , $f(x)$ approaches 0
  - This is written as $x \to \infty$, $f(x) \to 0$
- This is true for the negative x and y axis as well.

Example

The function f is defined by

$$\frac{1}{(4x+2)^2}, \quad x > -\frac{1}{2}$$

Find $\int_1^\infty f(x)\tilde{d}x$.

Answer

We begin by integrating

$$\int_1^\infty (4x+2)^{-2} = \left(\frac{(4x+2)^{-2+1}}{4 \times (-2+1)}\right)_1^\infty$$

Substitute a dummy variable, t, for the upper limit and evaluate the definite integral

$$\left(\frac{-1}{16t+8}\right) - \left(\frac{-1}{16(1)+8}\right)$$

As $t \to \infty$, $\frac{-1}{16t+8} \to 0$.

$$0 + \frac{1}{24}$$

So the answer to the improper integral is $\frac{1}{24}$.

Example 2

Evaluate whether the integral $\int_0^2 \frac{5}{x^2}\tilde{d}x$ is defined or undefined.

Answer

We begin by integrating

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$$\int_0^2 5x^{-2}\tilde{d}x = \left(\frac{5x^{-2+1}}{-2+1}\right)_0^2$$

Substitute a dummy variable, t, for the lower limit and evaluate the definite integral

$$\left(\frac{-5}{2}\right) - \left(-\frac{5}{t}\right)$$

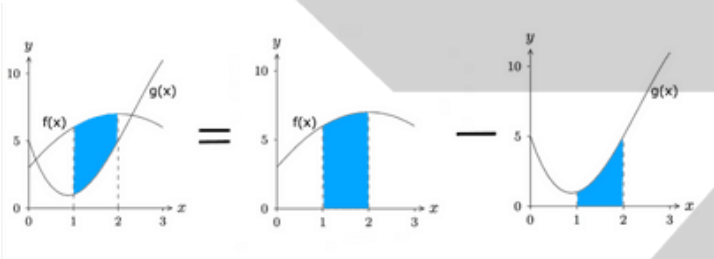
As $t \to 0$, $-\frac{5}{t} \to \infty$.

$$\frac{-5}{2} + \infty$$

As there is no finite value given to this improper integral, it is undefined.

1.8. Area between curves

- The area between two curves can be found by finding the difference of the areas of the top curve and bottom curve.



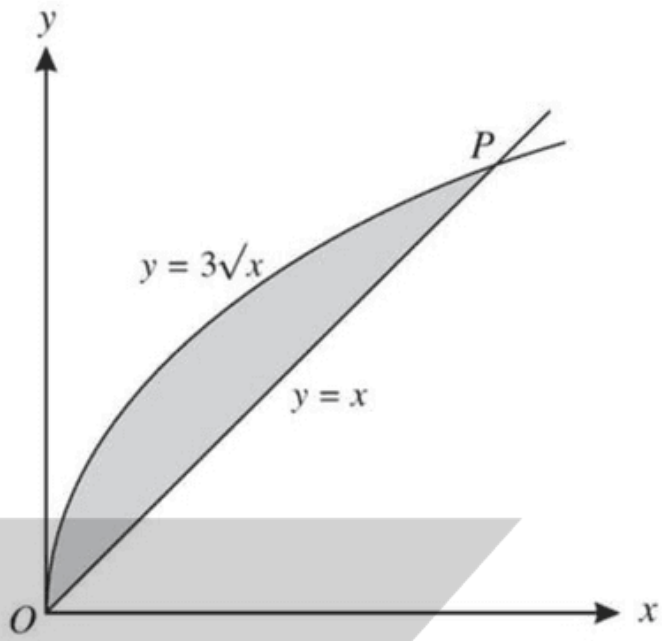
We can find the area between two curves as such

$$\int_a^b f(x) - g(x)\, dx \quad [\text{Where } a \le x \le b]$$

- While finding the area between two curves, it is important to identify which function is on top and on the bottom.
  - You may also need to find their points of intersection, and use them as the upper and lower limit in questions.

Example

The curve $y = 3\sqrt{x}$ and the line $y = x$ intersect at the origin O and the point P.



Find the area of the shaded region.

Answer

We must first find the points of intersection, to use them as our limits. The point O has coordinates $(0, 0)$. To find the x-coordinate of the point P, we must equate both curves.

$$3\sqrt{x} = x \implies x = 9$$

The point P has coordinates $(9, 9)$.

Method 1

Next we notice that curve $y = 3\sqrt{x}$ is above the line $y = x$. So we subtract the area of the line from the curve.

$$\int_0^9 3\sqrt{x}\, dx - \int_0^9 x\, dx = \int_0^9 3\sqrt{x} - x\, dx$$

$$\left(2x^{\frac{3}{2}} - \frac{x^2}{2}\right)_0^9$$

$$0 - \left(2(9)^{\frac{3}{2}} - \frac{9^2}{2}\right) = 13.5$$

The area between the two curves is 13.5.

Method 2

Next we notice that curve $y = 3\sqrt{x}$ is above the line $y = x$. So we subtract the area of the line from the curve. The line is a triangle with base length 9 and height 9.

$$\int_0^9 3\sqrt{x}\, dx - \left(\frac{1}{2} \times 9 \times 9\right)$$

$$(2x^{\frac{3}{2}})_0^9 - \left(\frac{81}{2}\right) = 2(9)^{\frac{3}{2}} - \frac{81}{2} = 13.5$$

The area between the two curves is 13.5.

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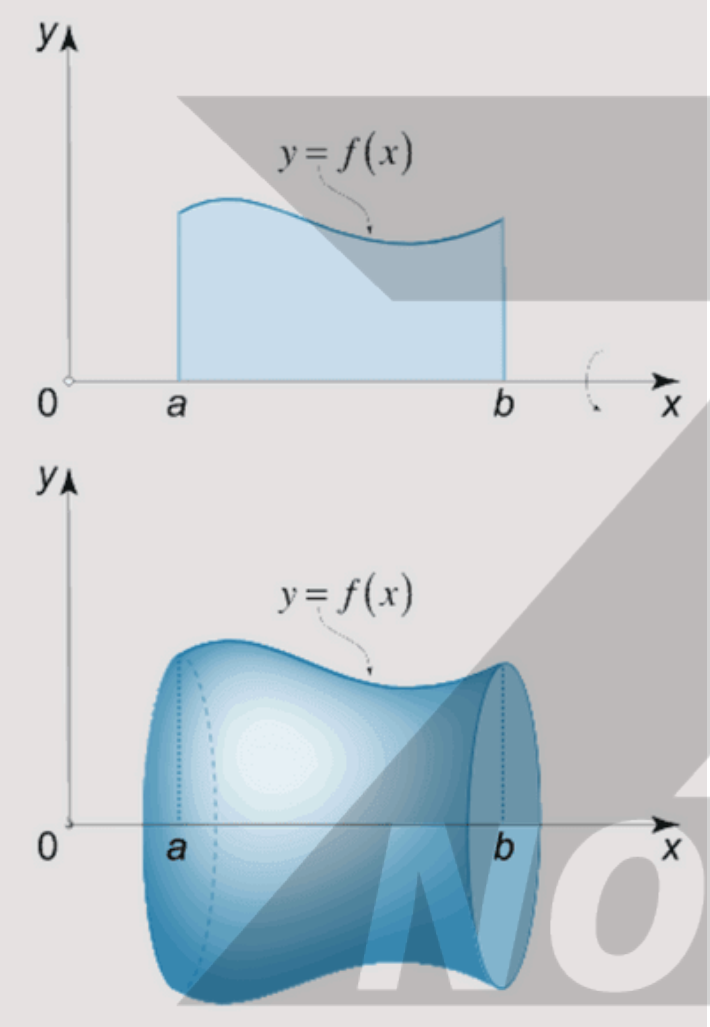
1.9. Volume of Revolution

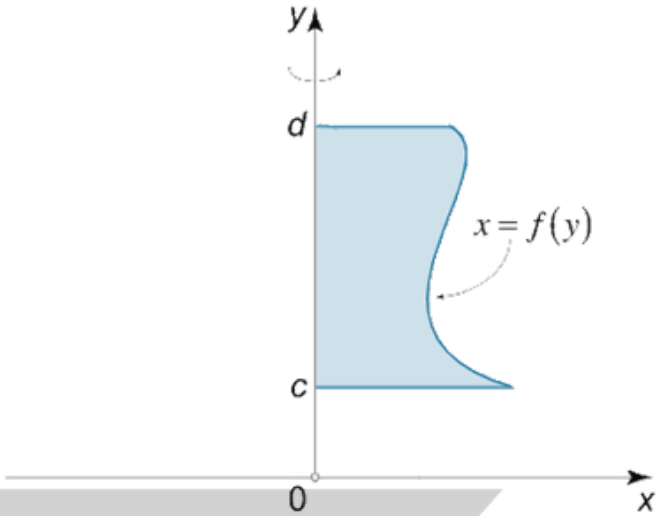
- The volume of revolution of a solid rotated around an axis by 360° degrees gives us a 3D shape.
- It is also possible to find the volume between two solid areas by subtracting the upper volume from the lower volume.
- To find the volume of revolution around the x-axis by 360 degrees

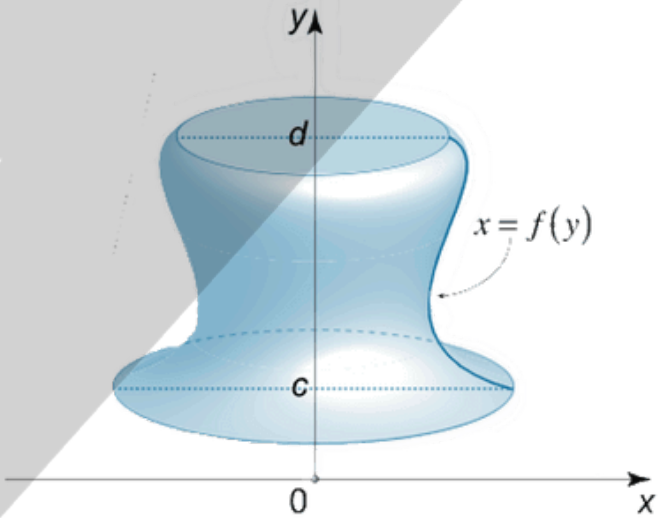
$$V = \pi \int_{x=a}^{x=b} y^2 \, dx$$



- To find the volume of revolution around the y-axis by 360 degrees

$$V = \pi \int_{y=c}^{y=d} x^2 \, dy$$





- Note that you do not need to square the equation if it is already in terms of y^2 or x^2

Example 1

Find the volume of the solid of revolution formed when the area under the graph of $y = x^3$ between x-coordinates 3 and 6 rotates 360° around the x-axis

Answer

First we need to get the equation in terms of y^2

$$y = x^3 \implies y^2 = x^6$$

Now we can evaluate the integral using the limits given

$$V = \pi \int_3^6 x^6 \, dx$$

$$\pi \left(\frac{x^{6+1}}{6+1} \right)_3^6$$

$$V = \pi \left(\frac{6^7}{7} - \frac{3^7}{7} \right)$$

$$V = \frac{277749\pi}{7}$$

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So the volume of revolution the around x-axis by 360 degrees is $\frac{277749\pi}{7}$

Example 2

Find the volume of revolution when the area under the graph of $y^4 = x^2$ between y-coordinates 2 and 4 rotated 360° around the y-axis.

Answer

As the equation is already in terms of x^2, we can directly evaluate the integral.

$$V = \pi \int_2^4 y^4 \, dy$$

$$\pi \left(\frac{y^{4+1}}{4+1} \right)_2^4$$

$$\pi \left(\frac{4^5}{5} - \frac{2^5}{5} \right)$$

$$V = \frac{992\pi}{5}$$

So the volume is equal to $\frac{992\pi}{5}$

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